

Table 1
Overall architecture of MaSBSLCNet.

Layer position	Layer details	Output shape	Param
Conv Stem	Conv2d	$64 \times 64 \times 16$	144
	BatchNorm2d + ReLU	$64 \times 64 \times 16$	32
BaseBlock 1 (stride=2)	Conv2d	$32 \times 32 \times 16$	144
	BatchNorm2d	$32 \times 32 \times 16$	32
	Conv2d	$32 \times 32 \times 16$	256
	BatchNorm2d + ReLU	$32 \times 32 \times 16$	32
	Conv2d	$64 \times 64 \times 16$	256
	BatchNorm2d + ReLU	$64 \times 64 \times 16$	32
	Conv2d	$32 \times 32 \times 16$	144
	BatchNorm2d	$32 \times 32 \times 16$	32
	Conv2d	$32 \times 32 \times 16$	256
	BatchNorm2d + ReLU	$32 \times 32 \times 16$	32
	InvertedResidual	$32 \times 32 \times 32$	0
BaseBlock 2 (stride=1)	Conv2d	$32 \times 32 \times 16$	256
	BatchNorm2d + ReLU	$32 \times 32 \times 16$	32
	Conv2d	$32 \times 32 \times 16$	144
	BatchNorm2d	$32 \times 32 \times 16$	32
	Conv2d	$32 \times 32 \times 16$	256
	BatchNorm2d + ReLU	$32 \times 32 \times 16$	32
	InvertedResidual	$32 \times 32 \times 32$	0
BaseBlock 3 (stride=2)	Conv2d	$16 \times 16 \times 32$	288
	BatchNorm2d	$16 \times 16 \times 32$	64
	Conv2d	$16 \times 16 \times 32$	1024
	BatchNorm2d + ReLU	$16 \times 16 \times 32$	64
	Conv2d	$32 \times 32 \times 32$	1024
	BatchNorm2d + ReLU	$32 \times 32 \times 32$	64
	Conv2d	$16 \times 16 \times 32$	288
	BatchNorm2d	$16 \times 16 \times 32$	64
	Conv2d	$16 \times 16 \times 32$	1024
	BatchNorm2d + ReLU	$16 \times 16 \times 32$	64
	InvertedResidual	$16 \times 16 \times 64$	0
BaseBlock 4 (stride=1)	Conv2d	$16 \times 16 \times 32$	1024
	BatchNorm2d + ReLU	$16 \times 16 \times 32$	64
	Conv2d	$16 \times 16 \times 32$	288
	BatchNorm2d	$16 \times 16 \times 32$	64
	Conv2d	$16 \times 16 \times 32$	1024
	BatchNorm2d + ReLU	$16 \times 16 \times 32$	64
	InvertedResidual	$16 \times 16 \times 64$	0
BaseBlock 5 (stride=2)	Conv2d	$8 \times 8 \times 64$	576
	BatchNorm2d	$8 \times 8 \times 64$	128
	Conv2d	$8 \times 8 \times 64$	4096
	BatchNorm2d + ReLU	$8 \times 8 \times 64$	128
	Conv2d	$16 \times 16 \times 64$	4096
	BatchNorm2d + ReLU	$16 \times 16 \times 64$	128
	Conv2d	$8 \times 8 \times 64$	576
	BatchNorm2d	$8 \times 8 \times 64$	128
	Conv2d	$8 \times 8 \times 64$	4096
	BatchNorm2d + ReLU	$8 \times 8 \times 64$	128
	InvertedResidual	$8 \times 8 \times 128$	0
BaseBlock 6 (stride=1)	Conv2d	$8 \times 8 \times 64$	4096
	BatchNorm2d + ReLU	$8 \times 8 \times 64$	128
	Conv2d	$8 \times 8 \times 64$	576
	BatchNorm2d	$8 \times 8 \times 64$	128
	Conv2d	$8 \times 8 \times 64$	4096
	BatchNorm2d + ReLU	$8 \times 8 \times 64$	128
	InvertedResidual	$8 \times 8 \times 128$	0
Pooling	AdaptiveAvgPool2d	$4 \times 4 \times 128$	0
Classifier	Flatten	2048	0
	BatchNorm1d	2048	4096
	Dropout	2048	0
	Linear	9	18441

A. Overall architecture of MaSBSLCNet

Table 1 details the architecture of MaSBSLCNet.